Lemma. Let V be a finite dimensional vector space over F.
If the characteristic polynomial of a linear operator $T: V \to V$ splits over F,
then there exists a basis β s.t $[T]_\beta$ is upper-triangle matrix.

Proof. put $n = \dim V$. Using mathematical induction for n. If $n=1$, trivial.
Suppose that for $n > 1$, the statement holds for every spaces having dimension as $n-1$.
Let V be a vector space with $\dim V = n$, and let $T: V \to V$ be a linear.
Since the characteristic polynomial of T splits over F, by assumption,
there exists a eigenvalue $\lambda_1 \in F$ corresponding to $v_1 \in V$.
By the Replacement Thm, we obtain a basis for V containing v_1

$$\{v_1, v_2, \cdots, v_n\}.$$

Denote $W_1 = \langle v_1 \rangle$, $W_2 = \langle v_2, \cdots, v_n \rangle$. Clearly, $V = W_1 \oplus W_2$.
observe that: W_1 is T-invariant but W_2 is not. Define two projections

$$P_1 : V \to V : x_1 + x_2 \mapsto x_1 \quad \text{and} \quad P_2 : V \to V : x_1 + x_2 \mapsto x_2. \quad (x_1 \in W_1, x_2 \in W_2)$$

Then, for any $x \in W_2$, $(P_2 \circ T)(x) \in W_2$, thus we can consider a restriction

$$P_2 \circ T|_{W_2} : W_2 \to W_2 : x \mapsto (P_2 \circ T)(x)$$

By the assumption, there exists a basis $\{w_2, \cdots w_n\}$ for W_2 s.t for $2 \leq j \leq n$,

$$(P_2 \circ T)(w_j) = \sum_{i=2}^{j} a_i \cdot w_i \quad \text{for some } a_i \in F.$$

Now, define for $1 \leq k \leq n-1$, $u_{k+1} = v_1 + w_{k+1}$ Then,
$\{v_1, u_2, \cdots, u_n\}$ is a basis for V, and satisfies for all $1 \leq j \leq n$,

$$T(v_1) = \lambda_1 v_1$$
$$T(u_2) = T(v_1 + w_2) = T(v_1) + T(w_2) = \lambda_1 v_1 + a_2 w_2$$
$$T(u_3) = T(v_1 + w_2 + w_3) = \lambda v_1 + a_2 w_2 + a_3 w_3$$
$$\vdots$$
$$T(u_n) = \lambda v_1 + a_2 w_2 + \cdots + a_n w_n.$$